Website Security Testing using Kali Linux Tools

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About the Project

This project focuses on testing the security of a website to identify vulnerabilities that attackers may exploit. A test website (in a legal and controlled environment) is analyzed using Kali Linux, which is a widely used operating system for ethical hacking and cybersecurity.

Web applications today are highly exposed to cyber threats due to improper coding practices, weak server configurations, and lack of regular security testing. This project aims to simulate real-world attack scenarios in a safe manner to understand how vulnerabilities are detected and mitigated.

By using tools like SQLmap and Nikto, this project demonstrates how security professionals identify risks and ensure that websites are protected from unauthorized access and attacks.

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Objectives of the Project

- To understand different types of web application vulnerabilities
- To perform basic penetration testing using Kali Linux tools
- To identify common attacks such as:
    - SQL Injection
    - Server misconfigurations
    - Unauthorized access risks
- To analyze vulnerabilities and generate reports
- To learn and apply basic protection techniques for securing websites

Scope of the Project

This project is limited to testing a sample or test website in a controlled environment. It focuses on identifying basic vulnerabilities and does not involve attacking real-world systems without permission. The scope includes vulnerability scanning, analysis, and suggesting preventive measures.

Tools Used

• **Kali Linux** – A Linux distribution designed for penetration testing and ethical hacking
• **SQL map** – An automated tool used to detect and exploit SQL Injection vulnerabilities
• **Nikto** – A web server scanner used to find outdated software, security issues, and misconfigurations

Working Principle

The project works on the principle of ethical hacking, where security testing is performed to identify weaknesses before attackers can exploit them.

The tools used simulate attack techniques and provide detailed reports about vulnerabilities present in the system.

Methodology

1. A test website URL is selected in a safe environment
2. The website is scanned using **Nikto** to detect:
    - o Outdated server software
    - o Security misconfigurations
    - o Known vulnerabilities
3. **SQLmap** is used to test whether the website is vulnerable to SQL Injection attacks
4. The scan results are carefully analyzed
5. Identified vulnerabilities are documented
6. Security recommendations are provided to improve protection

Results and Analysis

The scanning process successfully identified suspicious patterns and potential vulnerabilities in the test website. Some of the findings include:

• Detection of unusual inputs indicating possible SQL Injection risks
• Identification of server misconfigurations
• Detection of suspicious data samples indicating potential data poisoning

These results show that even simple vulnerabilities can pose serious risks if not addressed properly.

```
Session  Actions  Edit  View  Help
Total suspicious samples: 2

┌──(scanner_env)─(kali㉿kali)-[~/ai_poison_project]
└─$ nano forensic_scanner.py

┌──(scanner_env)─(kali㉿kali)-[~/ai_poison_project]
└─$ nano train_model.py

┌──(scanner_env)─(kali㉿kali)-[~/ai_poison_project]
└─$ python forensic_scanner.py

Scanning Dataset for Poisoning Attacks ...

⚠ Suspicious Training Samples:

                        text  anomaly
4   normal user login request       -1
8  admin access with backdoor       -1

Total suspicious samples: 2

┌──(scanner_env)─(kali㉿kali)-[~/ai_poison_project]
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Model Accuracy: 1.0

⚠️ Potential Data Poisoning Detected!

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Advantages of the Project

- Helps in identifying security weaknesses in websites
- Improves knowledge of ethical hacking techniques
- Provides practical exposure to cybersecurity tools
- Helps developers build secure applications
- Useful for learning real-world cyber attack scenarios

Limitations of the Project

• Limited to test environments only
• Cannot detect all types of advanced vulnerabilities
• Requires technical knowledge to interpret results
• Tools may produce false positives in some cases

Future Scope

• Use advanced tools like **Burp Suite** and **Metasploit**
• Implement automated security testing systems
• Integrate AI-based threat detection techniques
• Perform real-time monitoring of web applications
• Extend testing to mobile and cloud applications

Applications of the Project

• Helps developers design secure websites
• Useful in cybersecurity auditing and testing
• Creates awareness about secure coding practices
• Helps organizations prevent cyber attacks
• Useful for careers in Cyber Security, Networking, and Ethical Hacking

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Conclusion

This project provides hands-on experience in website security testing using Kali Linux tools. It highlights the importance of identifying vulnerabilities and performing regular security assessments to protect web applications from cyber threats and unauthorized access.

The project also emphasizes the role of ethical hacking in improving system security and preventing potential cyber attacks.